



WEEK 02 WHAT IS GIS

LECTURE

INSTRUCTOR: YANAN WU

FALL 2025

2.1

INTRODUCTION TO GIS

What is a GIS?

- Geographic Information
 - information about places on the earth's surface
 - knowledge about “what is where when”
- GIS--what's in the S?
 - Systems: the technology/tool
 - Science: the concepts and theory

ARE THEY TECHNOLOGY OR SCIENCE IN GIS?

- ArcGIS PRO
- Drone with camera
- Coordinate system and map projection
- Web Mapping tool
- Network Analysis

Do you think you have used GIS in your daily life?



DO THEY USE GIS?

- Fitness apps: mapping runs, biking paths
- Real estate apps: houses displayed on maps with neighborhood data
- Apple Find My iPhone
- Pokemon Go

GIS = data + location + analysis + visualization, not just seeing a map

GIS IN THE NEWS

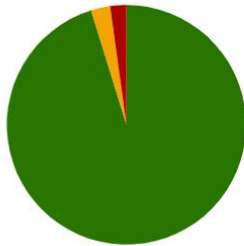
■ Natural Disaster

NCDOT Road Reopening for Hurricane Helene 2024

Start Date:
08/23/2025 and before

Total Incidents
1,507
since beginning of Storm

Total Closures
1,468
since beginning of Storm



■ Fully Open 1,435
■ Partial Access 39
■ Current Closed 33

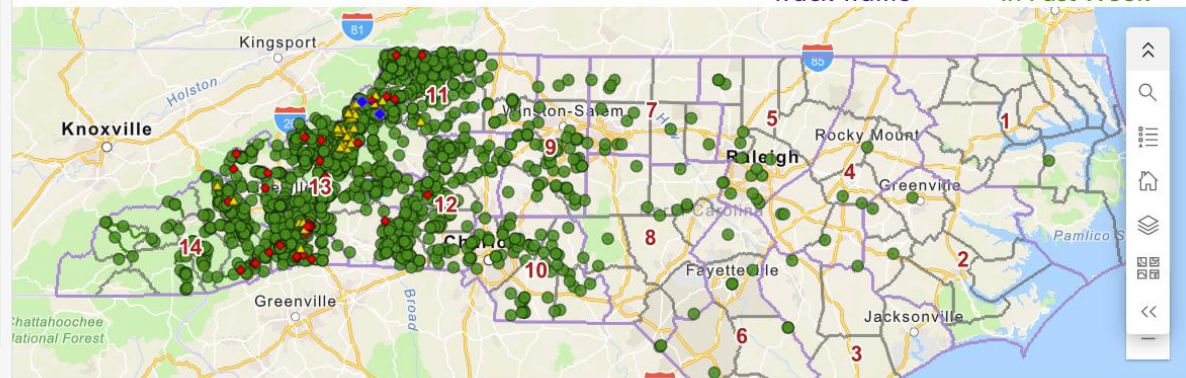
1,435
Fully Open

33
Current Closed

39
Partial Access

Closed to
1
Truck Traffic

Reopened
3
in Past Week



Esri, TomTom, Garmin, FAO, NOAA, USGS, EPA, NPS, USFWS | NCDOT GIS Unit

Powered by Esri

Incidents

Active & Cleared

Condition

Rt Type

The DriveNC/TIMS Incidents data contains the general location of and details about incidents that can affect travel on roads maintained by the North Carolina Department of Transportation.

All widgets are filtered by Event Name = 2024 Hurricane Helene & Created from Concurrent = False & StartDate is or is before Today

Fully Open: All incidents Expired or Cleared

Current Closed: Active Incidents - Ramp Closed Road Closed, Road Closed with Detour, Road Impassable

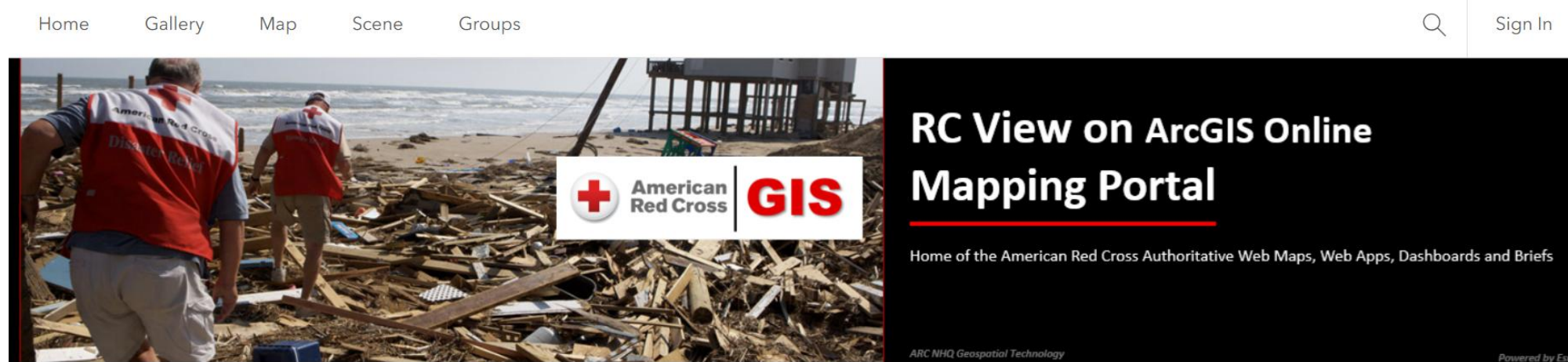
Partial Access: All Active incidents that are not Closed

Overview

Instructions

Active Closures List

■ Emergency Response and Recovery



Featured Content

**Tropical
Hazard &
Preparedness**

 **Web App**

**Weather
Hazard &
Preparedness**

 **Web App**

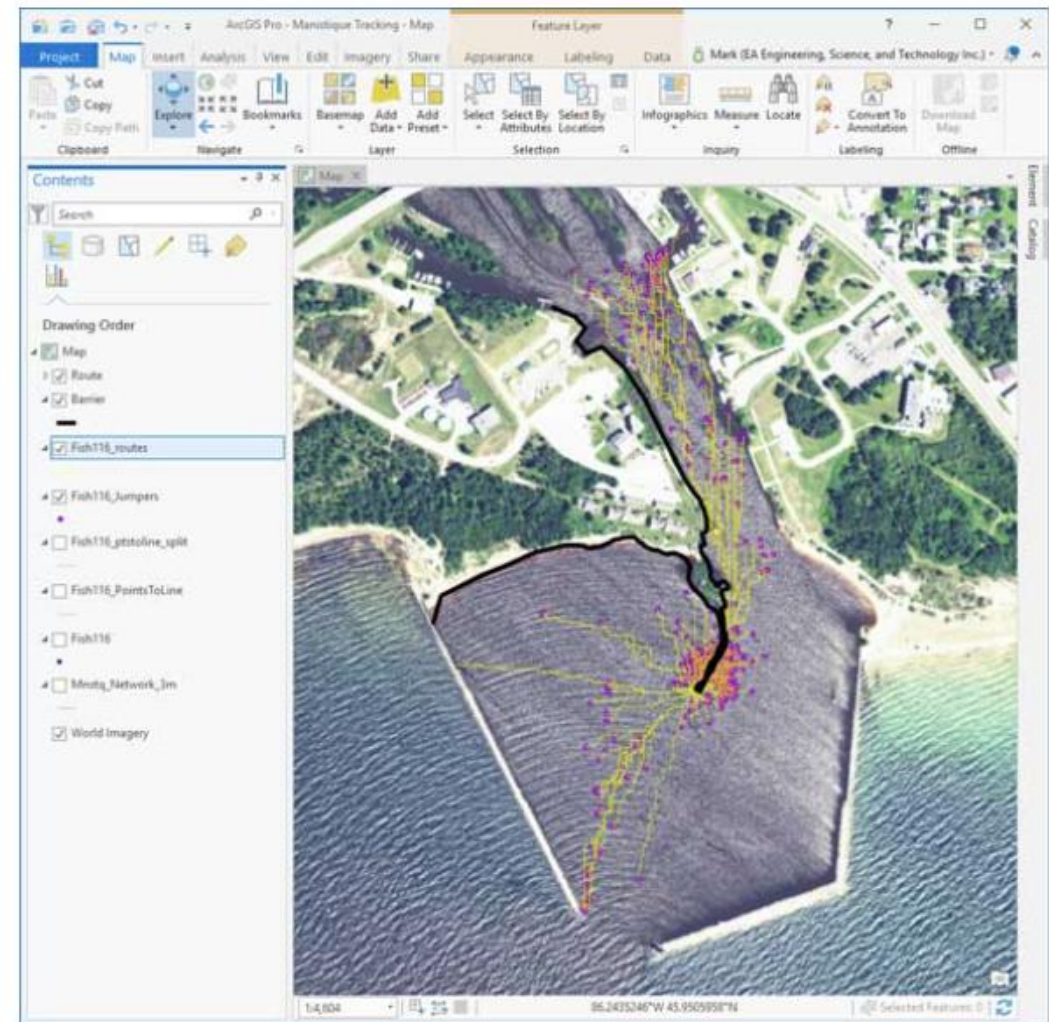
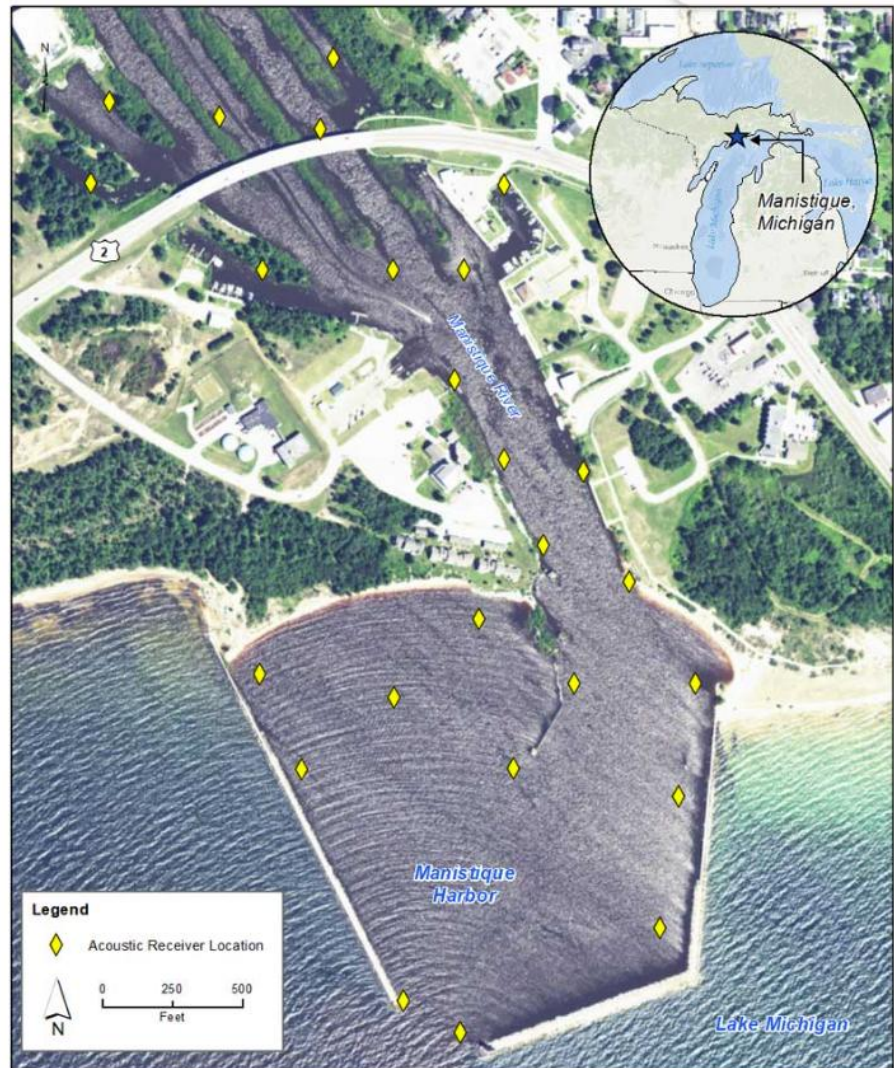
**Wildfires
Hazard &
Preparedness**

 **Web App**

**DRO DDA
Dashboards**

 **Gallery**

■ Tracking Fish

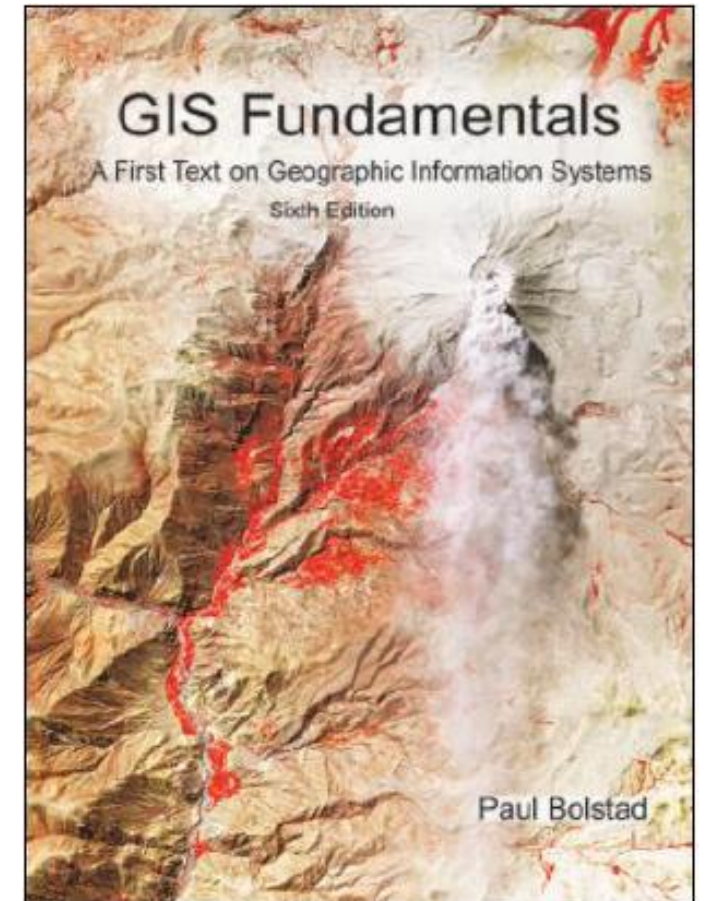


These revised line segments were generated by the Network Analyst extension.

GI SYSTEMS

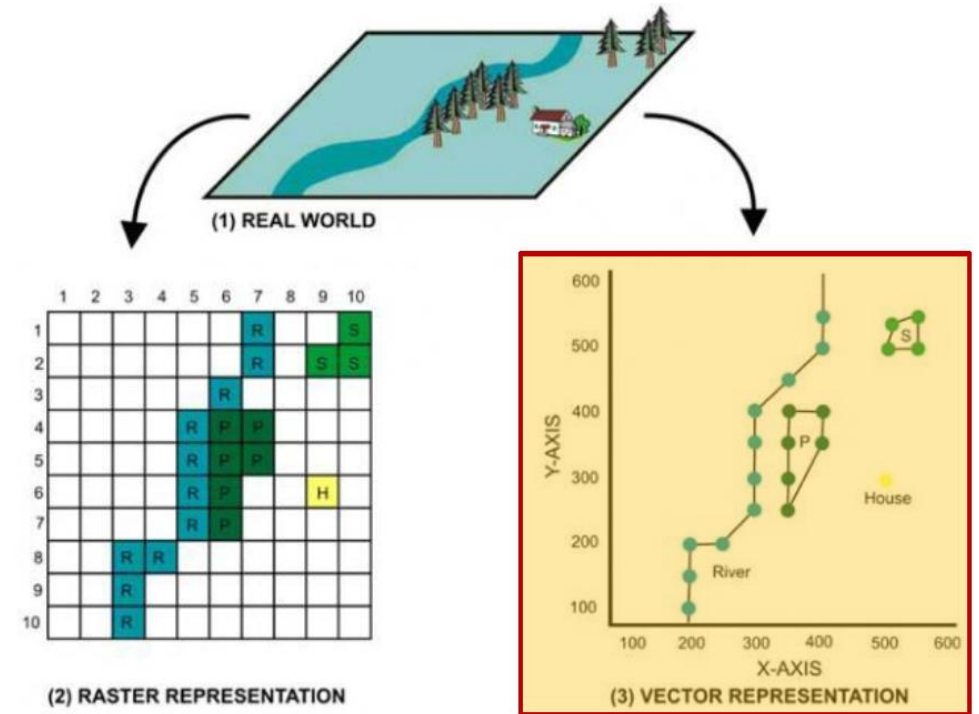
A computer-based system to aid in the collection, maintenance, storage, analysis, output, and distribution of spatial data and information

-Paul Bolstad



GI SCIENCE

Geographic information science is the development and use of theories, methods, technology, and data for understanding geographic processes, relationships, and patterns



<https://sqlserverrider.wordpress.com/tag/raster-graphics/>

- Goodchild, M. F. (1992).

GI technologies

3S

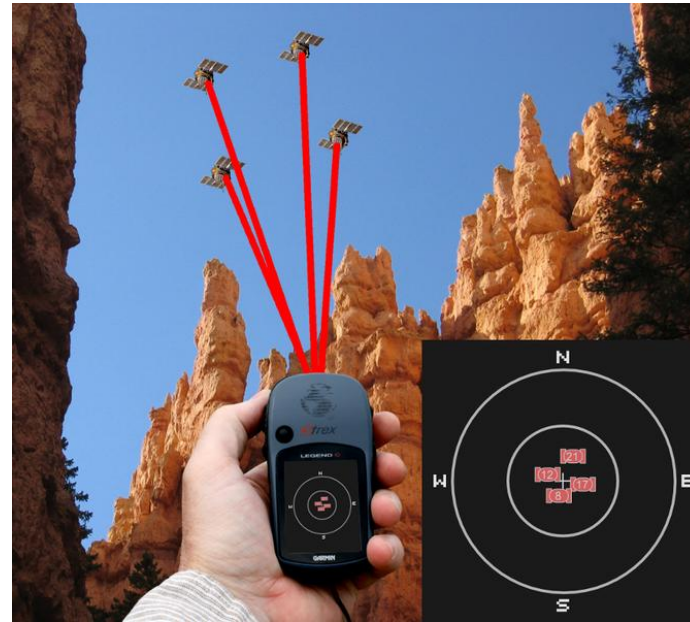
Global Positioning Systems

Remote Sensing RS



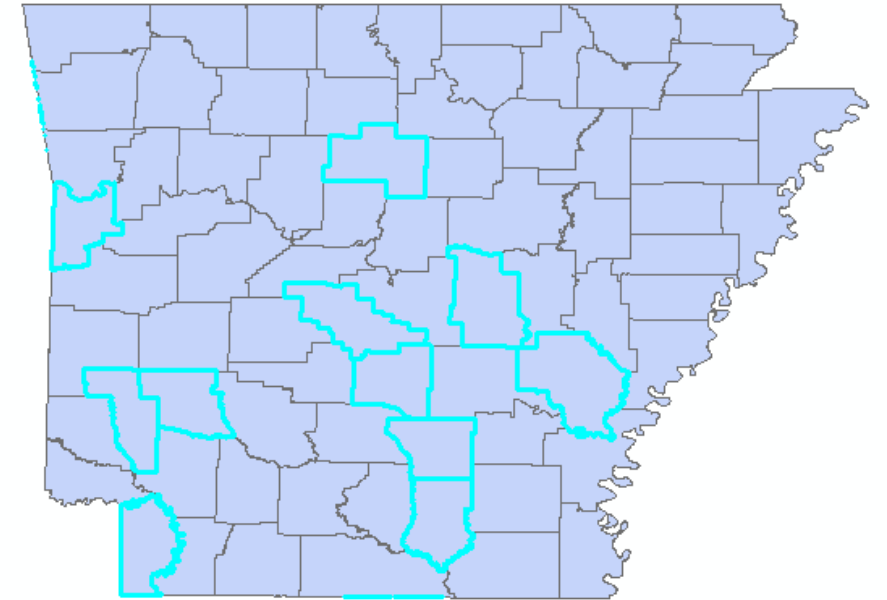
<https://landsat.usgs.gov/landsat-8>

GPS



https://commons.wikimedia.org/wiki/File:Bad_gps.png

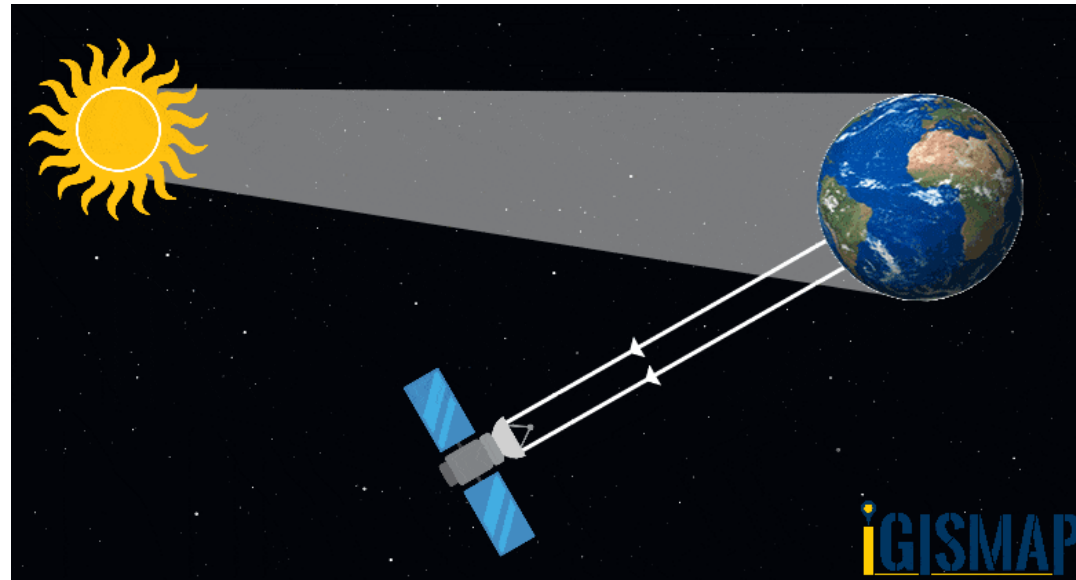
GIS



GI technologies

3S

- RS
 - use of satellites or aircraft to capture information about the earth's surface



■ Wildfire

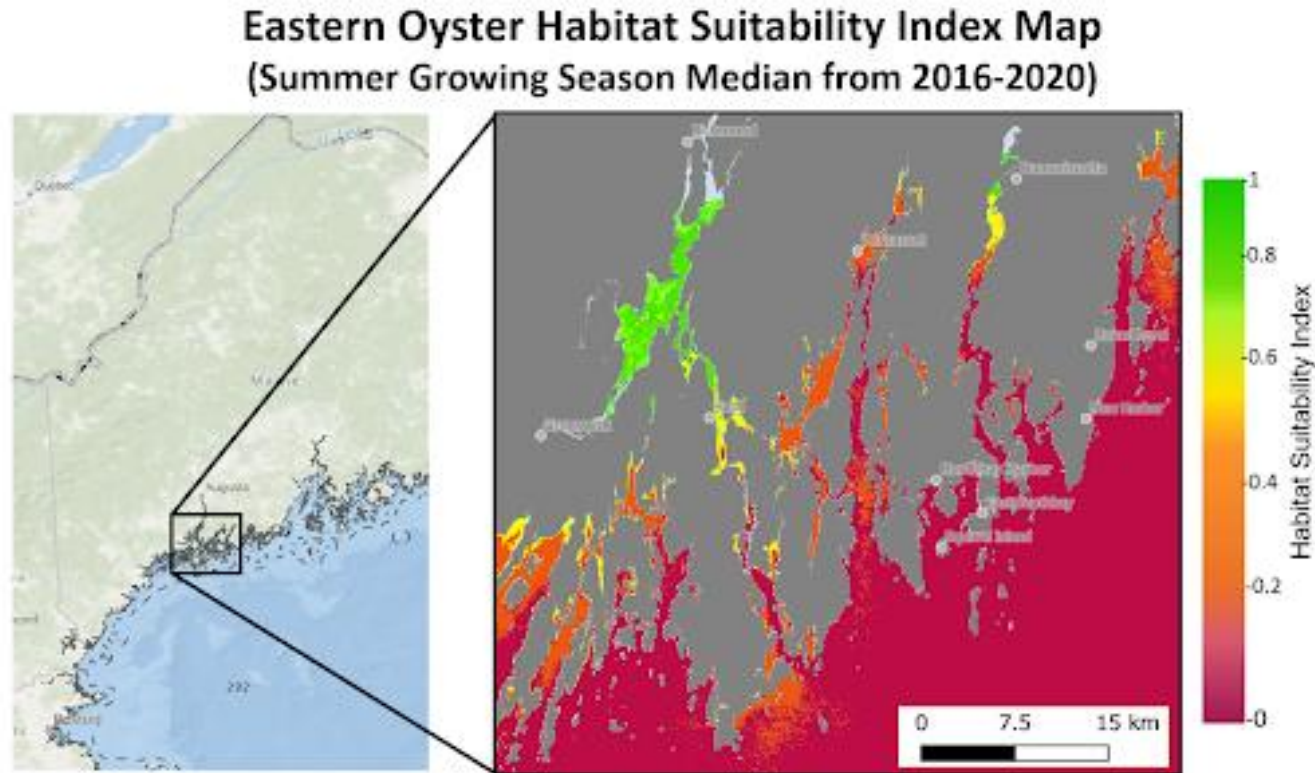


Blue Cut fire in California at 10:36 p.m. Pacific Daylight Time on August 17, 2016, as observed by the Thermal Infrared Sensor (TIRS) on Landsat 8 .

The image shows the amount of heat (thermal energy) radiating from the fiery landscape. Cooler areas are dark, while warmer areas are bright.

Image credit: Joshua Stevens, NASA Earth Observatory

■ Shellfish Suitability



The satellites measured sea surface temperature, plankton abundance, and water clarity.

Scientists are using satellite data to support the identification of sites most suitable for oyster farming in the Gulf of Maine (Credit: NOAA).

GI technologies

- GPS

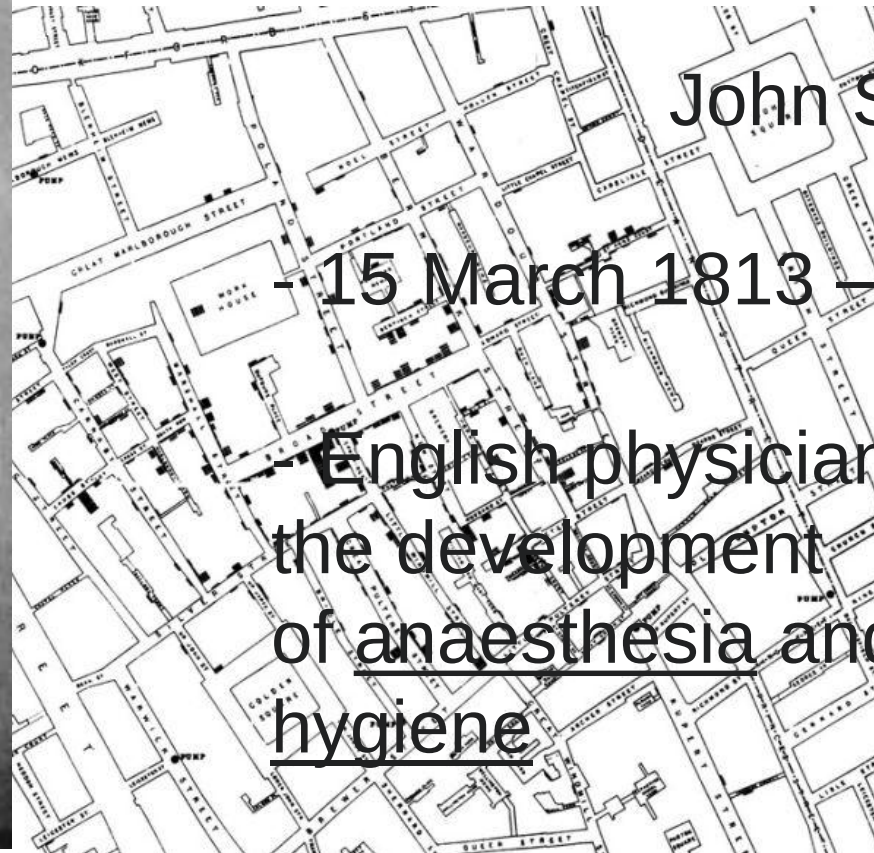
- a system of earth-orbiting satellites which can provide precise location on the earth's surface (in lat/long coordinates or equiv.)



GPS and RS are sources of input data for a GIS

PRECURSOR TO GIS

■ John Snow's Cholera Map

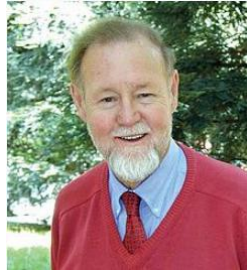


John Snow

- 15 March 1813 – 16 June 1858

- English physician and a leader in the development of anaesthesia and medical hygiene

HISTORY



1960s

Dr. Roger Tomlinson
Canada GIS



1969

Mr. Jack Dangermond

U.S. Census Bureau
Harvard Lab developed key
tools that advanced GIS

1970s

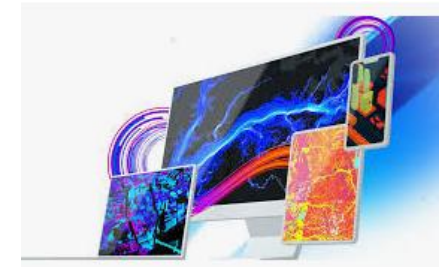


1970s



Arc/info, MapInfo

1980s



Desktop GIS

1990s



Web GIS

2000s

<https://www.esri.com/en-us/what-is-gis/history-of-gis>

https://en.wikipedia.org/wiki/Geographic_information_system

FUNCTIONS OF A GIS

1. Capture
2. Store
3. Query
4. Analyze
5. Display

Functions: *Capture*



Volunteered Geographic Information



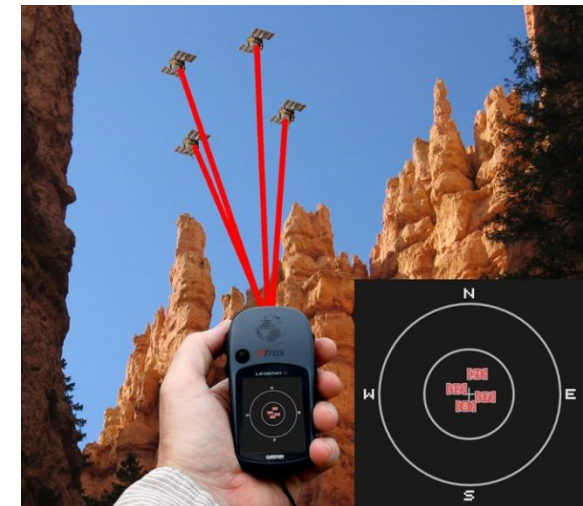
Mapathon



<https://landsat.usgs.gov/landsat-8>



https://commons.wikimedia.org/wiki/File:Intersect_UAV_B_3.1.png



https://commons.wikimedia.org/wiki/File:Bad_gdop.png

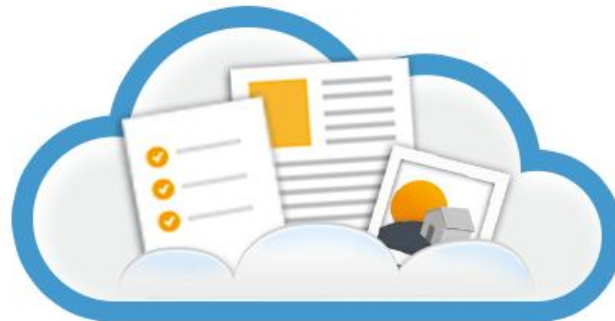
Functions: *Store*



Local disk

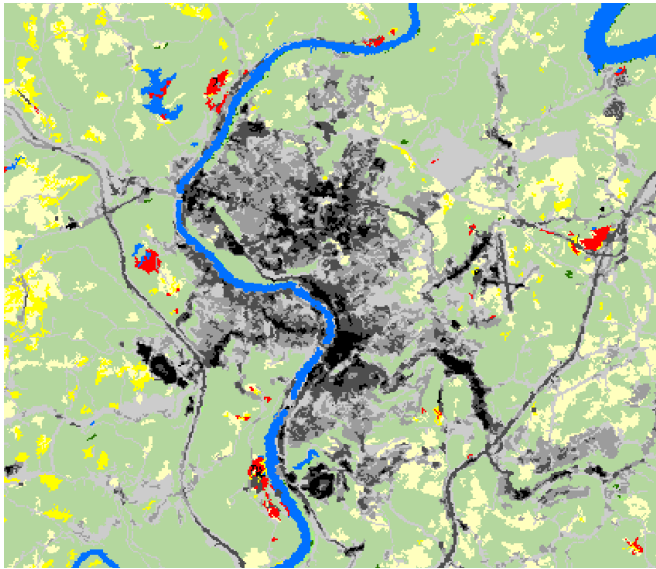


Server

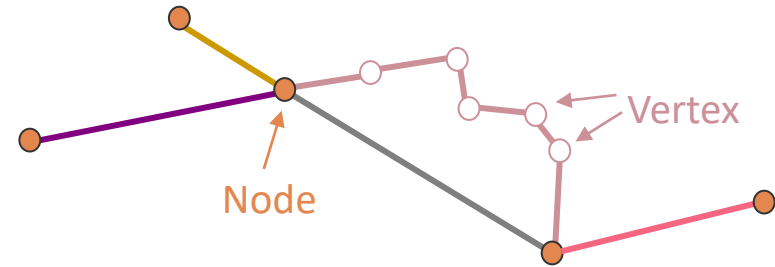


Cloud

Functions: *Store*



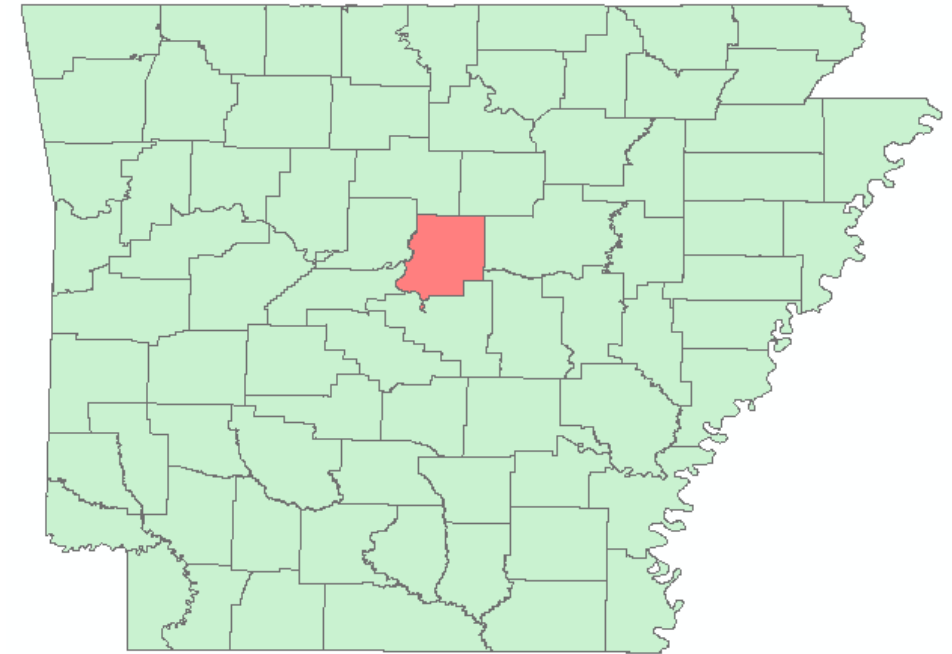
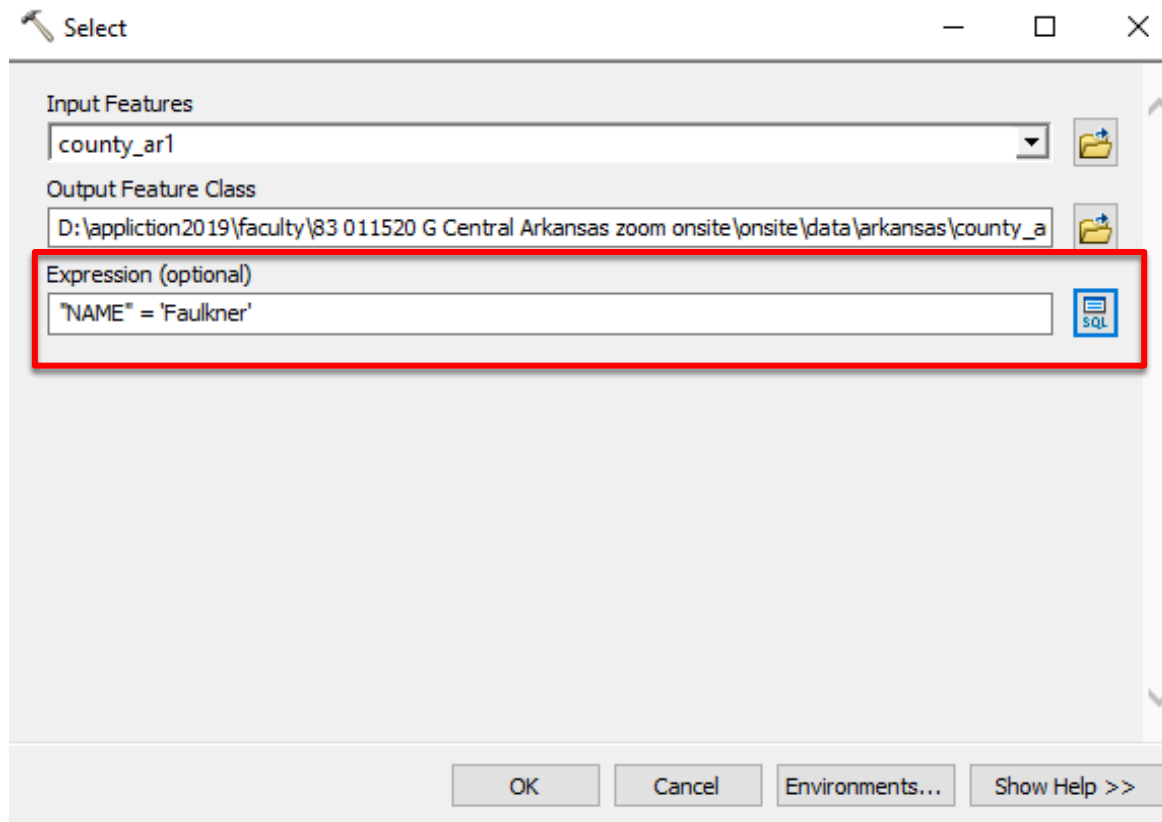
Raster



Vector

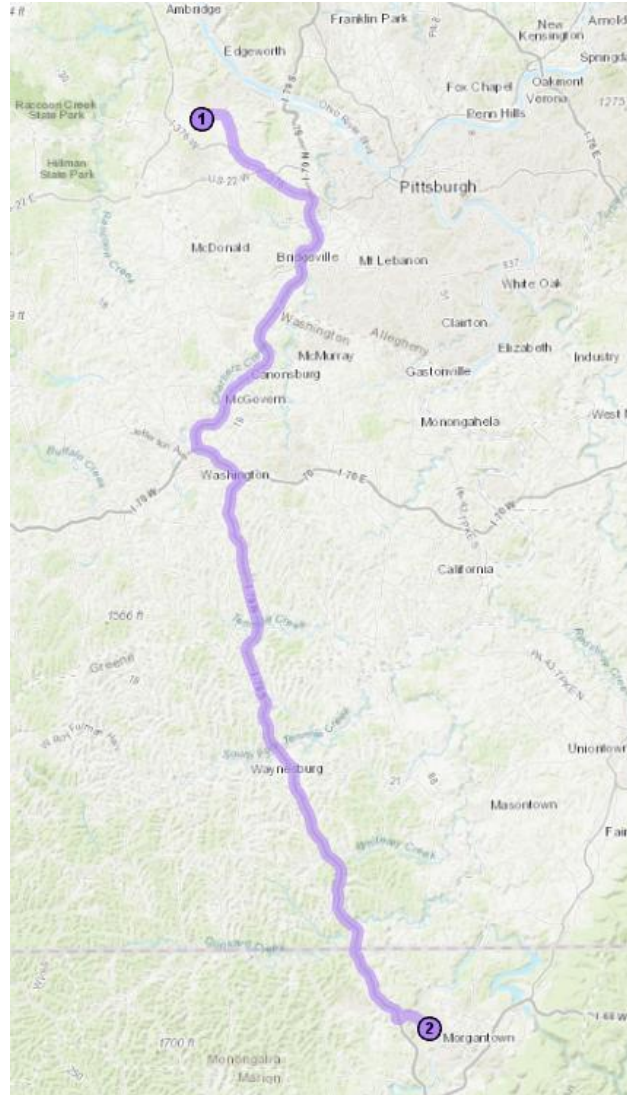
Functions: *Query*

Structured Query Language (SQL)

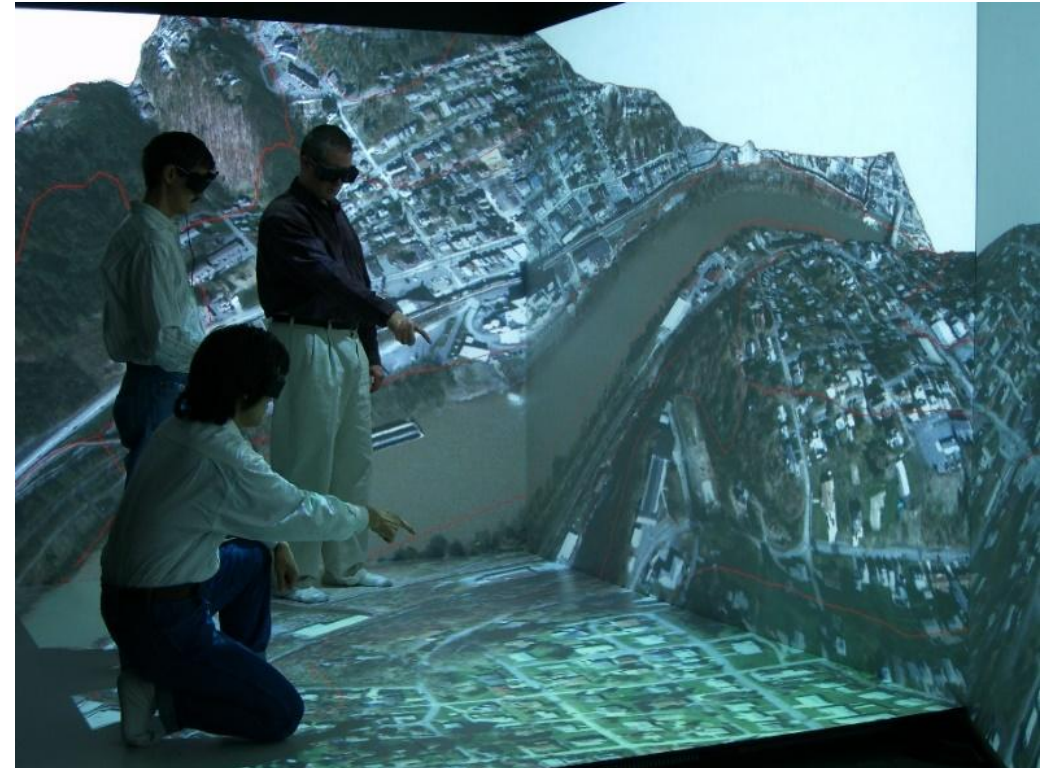
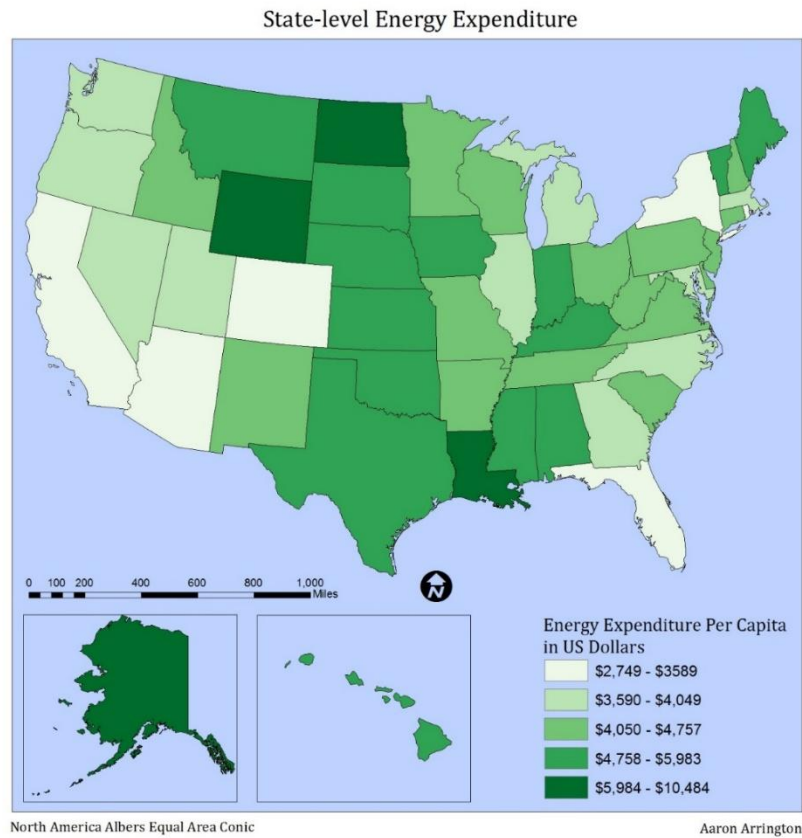


Functions: *Analyze*

1. Directions
2. Buffer
3. Proximity
4. Suitability
5. Likelihood
6. Best locations
7. Worst locations
8. Etc.



Functions: *Display*



Virtual Reality

Components of a GIS

1. People
2. Data
3. Hardware
4. Software
5. Methods/Protocol



https://www.rst2.org/ties/GENTOOLS/comp_gis.html

Components: *People*



<https://www.argisusers.org/>



<http://www.aag.org/>



<https://www.asprs.org/>



GIS Certification Institute

<https://www.gisci.org/>



esri

Academy

<https://www.esri.com/training/>

Components: *People*

○ Department of Geography @UCA

Programs

Undergraduate Majors

Geography

Geospatial Concentration

Environmental Science – Planning & Administration

Undergraduate Minors

Geography

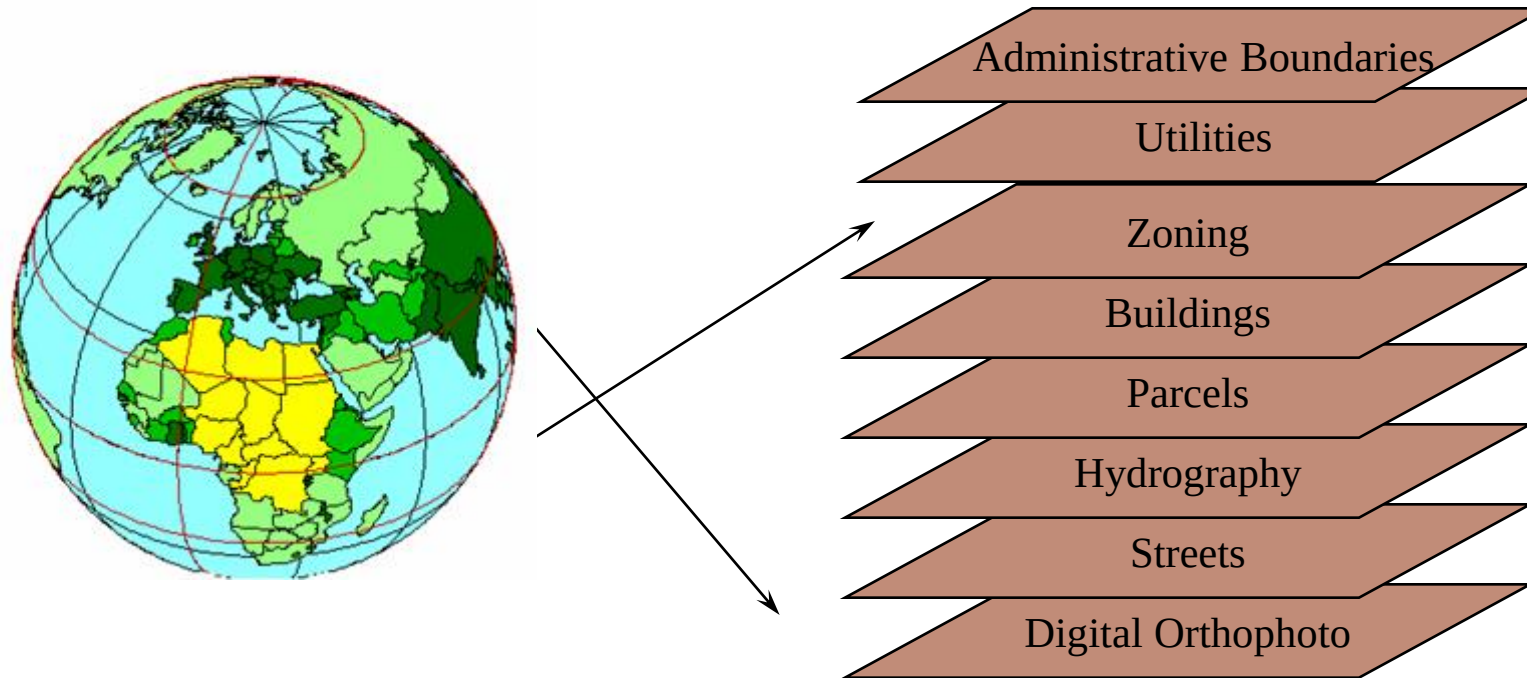
GIS

Graduate

Master of GIS

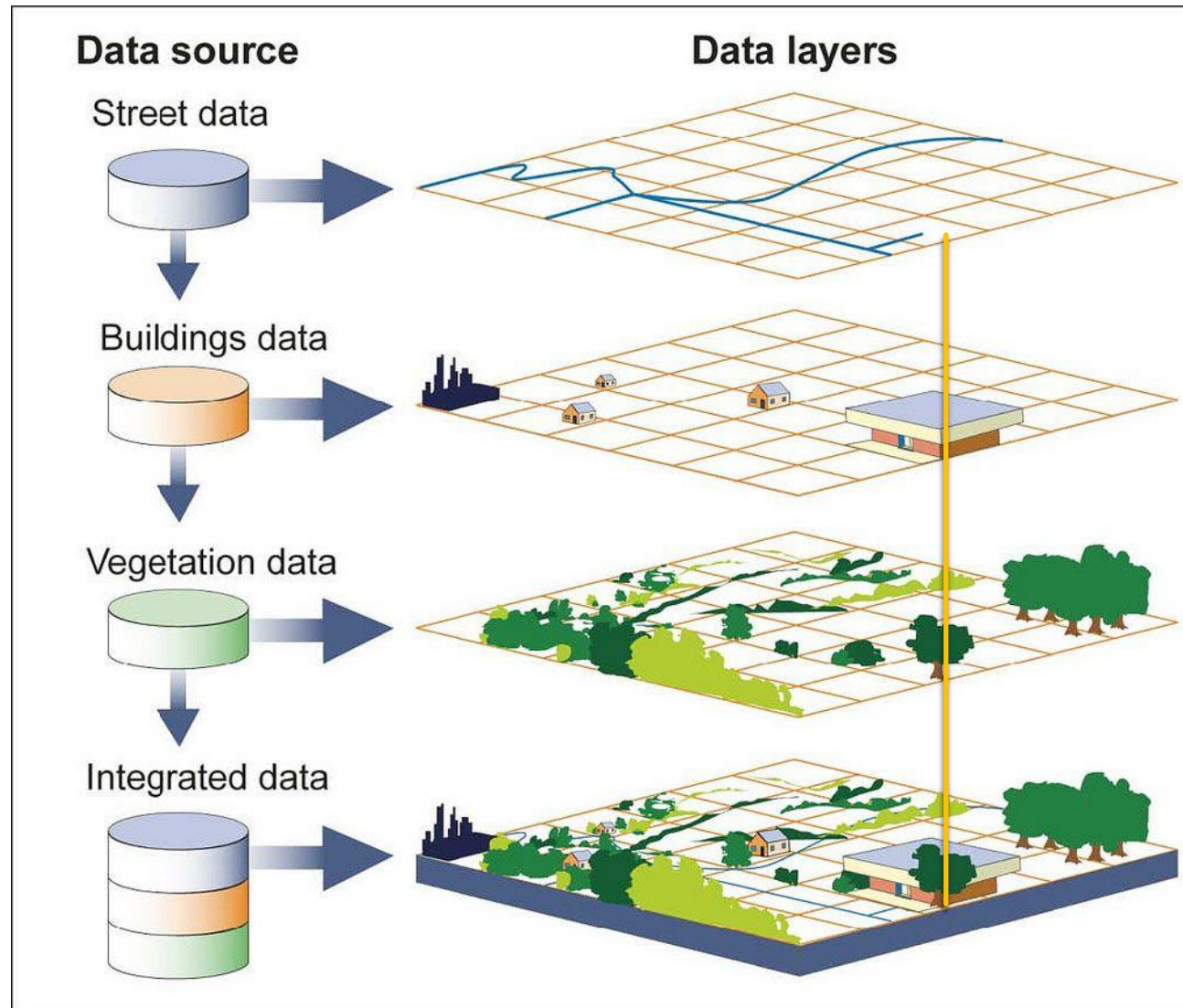
GIS Certificate

Components: *Data*



- Data is organized by layers, coverages or themes, *with each layer representing a common feature.*
- Layers are integrated using explicit location on the earth's surface, *thus geographic location is the organizing principal.*

Components: *Data*



Components: *software*



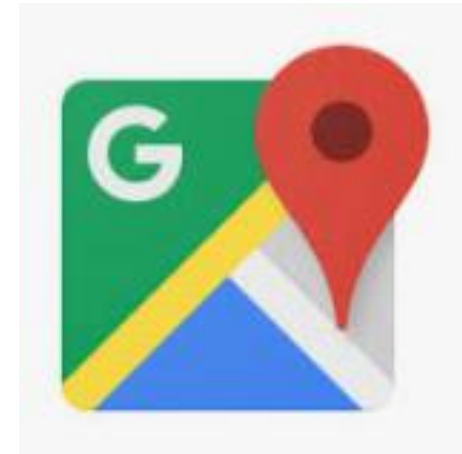
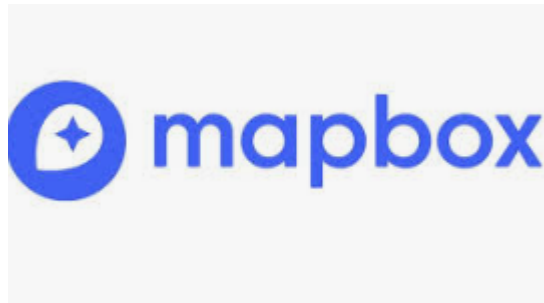
TYPES OF GIS & MAPPING SOFTWARE

Type	Analysis Power	Example(s)
Geobrowser	Weak mainly only to display data)	Google Maps, Google Earth, Apple Maps, Waze, etc.



TYPES OF GIS & MAPPING SOFTWARE

Type	Analysis Power	Example(s)
Web-based	Medium able to upload additional data, customize display, and perform basic analyses)	Carto, ArcGIS Online, Mapbox, Google MyMaps, etc.



TYPES OF GIS & MAPPING SOFTWARE

Type	Analysis Power	Example(s)
Desktop	Strong installed locally, provides full control of map creation, and perform advanced analyses)	ArcGIS Pro, QGIS



Which desktop software should we use today?

ArcGIS Pro

1. Commercial (expensive, requires license)
2. Windows only
3. Powerful, but heavy on computer resources
4. Strong integration with ArcGIS Online
5. Excellent training & official support
6. Full professional toolkit

QGIS

1. Free, open-source
2. Works on Windows, Mac, Linux
3. Lightweight, runs smoothly on most computers
4. Many tools available (though some functions less polished)
5. Community tutorials & forum support
6. Flexible (anyone can develop tools, but documentation varies)

Components: *hardware*



https://commons.wikimedia.org/wiki/File:Intersect_UAV_B_3.1.png



https://www.microsoft.com/en-us/p/surface-laptop-3/8VFGGH1R94TM/F29J?activetab=overview&source=googleshopping&OCID=AID2000022_SEM_mRUxj5Qp

<http://geospatialfieldmethodstanner.blogspot.com/2014/04/field-activity-9-surveying-with-topcon.html>



Components: *Method*

- ✓ Table Query
- ✓ Map design
- ✓ Digitizing
- ✓ Georeferencing
- ✓ Resampling
- ✓ Vector spatial analysis
- ✓ Raster spatial analysis
- ✓ Digital terrain analysis
- ✓ Geospatial modeling
- ✓ ArcGIS Model Builder
- ✓ Weighted overlay
- ✓ Spatial statistical analysis
- ✓ Spatial interpolation

- [-]  Spatial Statistics Tools
 - [-]  Analyzing Patterns
 -  Average Nearest Neighbor
 -  High/Low Clustering (Getis-Ord General G)
 -  Incremental Spatial Autocorrelation
 -  Multi-Distance Spatial Cluster Analysis (Ripleys K Function)
 -  Spatial Autocorrelation (Morans I)
 - [-]  Mapping Clusters
 -  Cluster and Outlier Analysis (Anselin Local Morans I)
 -  Grouping Analysis
 -  Hot Spot Analysis (Getis-Ord Gi*)
 -  Optimized Hot Spot Analysis
 -  Similarity Search
 - [-]  Measuring Geographic Distributions
 -  Central Feature
 -  Directional Distribution (Standard Deviational Ellipse)
 -  Linear Directional Mean
 -  Mean Center
 -  Median Center
 -  Standard Distance



GIS

MORE THAN MAPS

Applications

- **Urban Planning, Management & Policy**

- Land acquisition
- Housing renovation programs
- Emergency response
- Tax assessment

- **Environmental Sciences**

- Monitoring environmental risk
- Modeling stormwater runoff
- Management of watersheds, floodplains, wetlands, forests, aquifers
- Groundwater modeling and contamination tracking

- **Political Science**

- Redistricting
- Analysis of election results
- Predictive modeling

Applications

- **Civil Engineering/Utility**

- Locating underground facilities
- Designing alignment for freeways, transit
- Coordination of infrastructure maintenance

- **Business**

- Demographic Analysis
- Market Penetration/ Share Analysis
- Site Selection

- **Education Administration**

- Attendance Area Maintenance
- Enrollment Projections
- School Bus Routing

- **Real Estate**

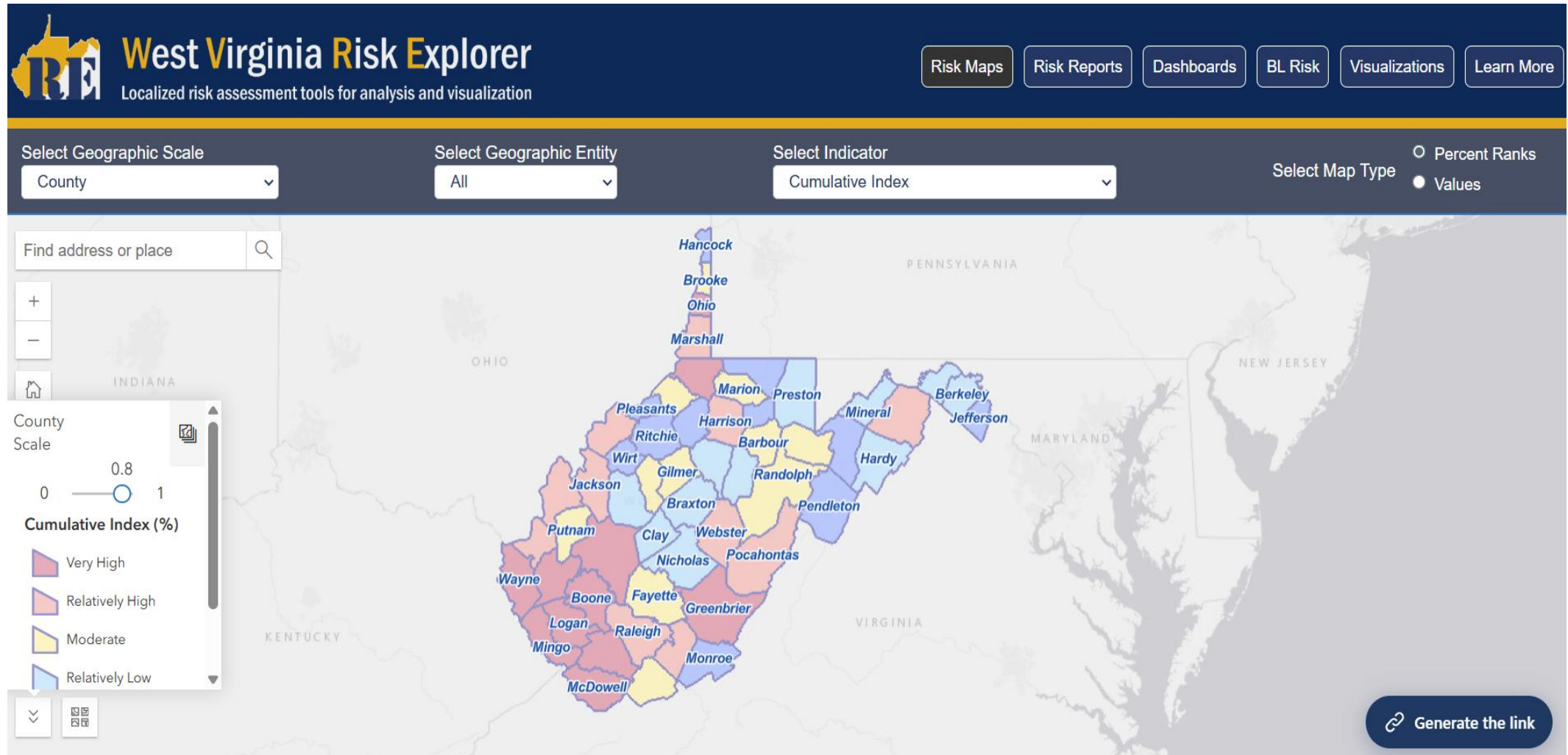
- Neighborhood land prices
- Traffic Impact Analysis
- Determination of Highest and Best Use

- **Health Care**

- Epidemiology
- Needs Analysis
- Service Inventory

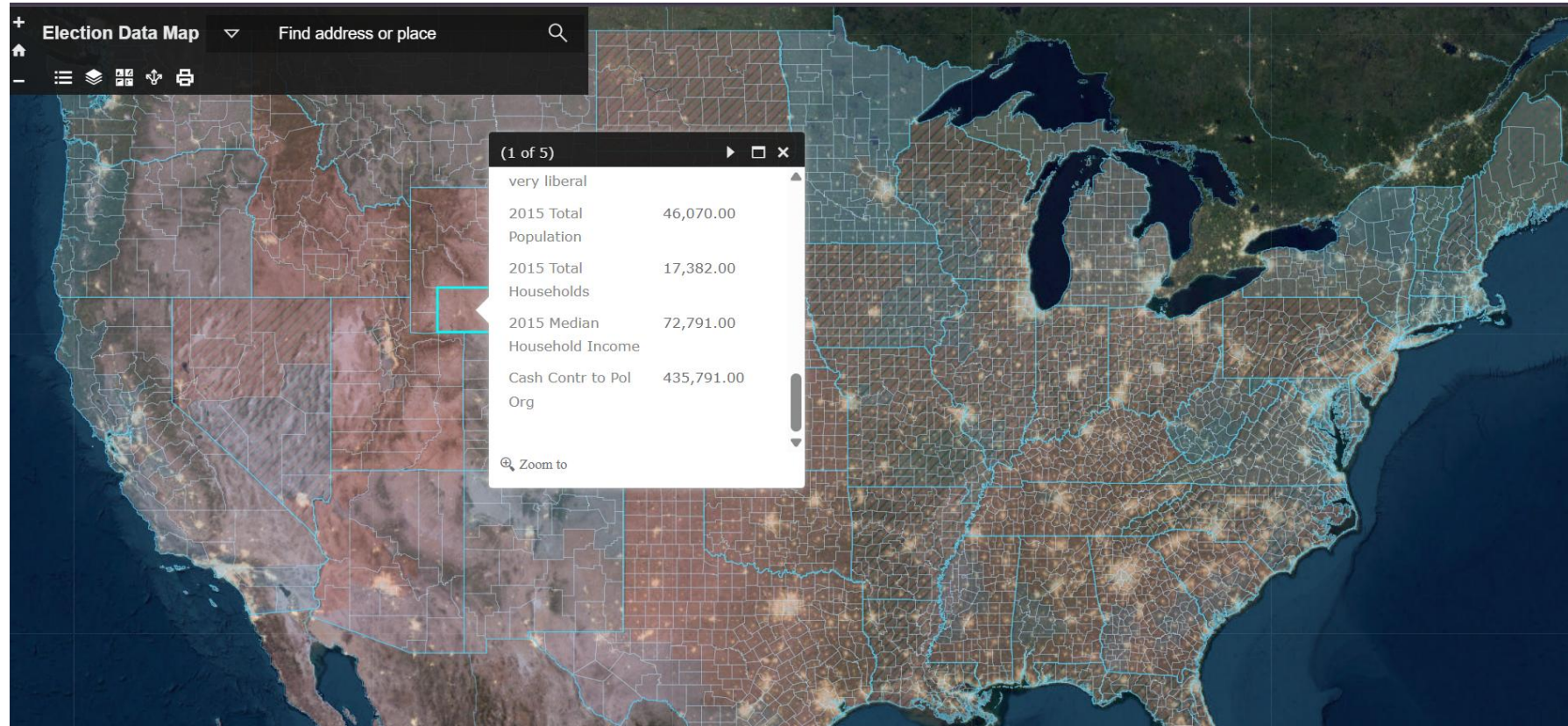
Example: *GIS in Environmental Science*

West Virginia Risk Explorer



Example: *GIS in Political Science*

2020 Presidential Election



Example: *GIS in Health Care*

Traveling during early stage of COVID-19 period



<https://geods.geography.wisc.edu/>



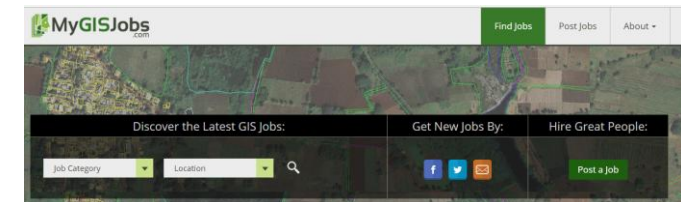
GIS Application in Modern World

GISer Job Market

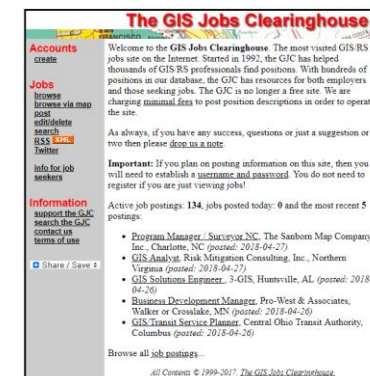
- **Local Government**
 - Planning and environmental management
 - property records and appraisal
- **Real Estate and Marketing**
 - Retail site selection, site evaluation
- **Public safety and defense**
 - Crime analysis, fire prevention, emergency management, military/defense
- **Natural resource exploration/extraction**
 - Petroleum, minerals, quarrying
- **Transportation**
 - Airline route planning, transportation planning/modeling
- **Public health and epidemiology**
- **The Geospatial Industry**
 - Data development, application development, programming



<https://www.gisjobs.com/>



<https://www.mygisjobs.com/>

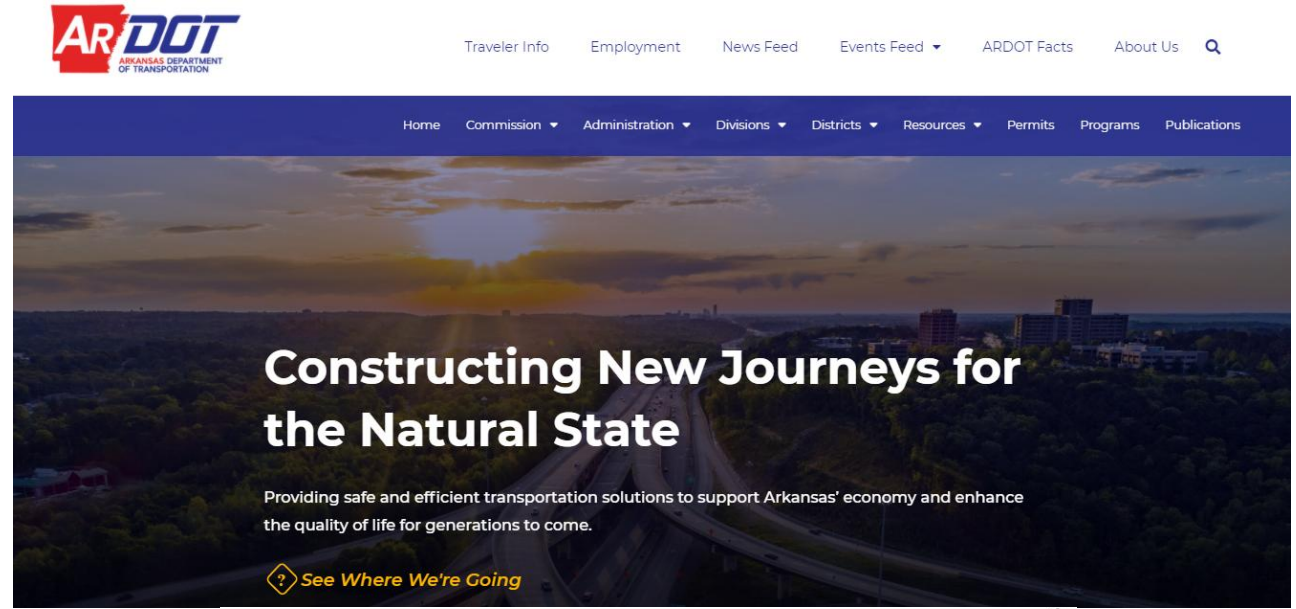


<http://www.gjc.org/>

Geography Students placement @UCA

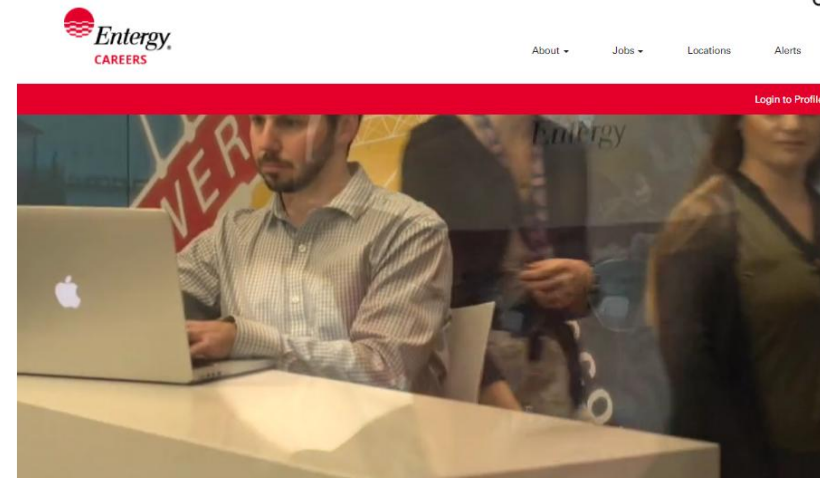
2020-2021

1. Four students worked as full time at ARDOT
2. Three students in master GIS program @UCA (One student is supported by Arkansas Space Grant Consortium project)



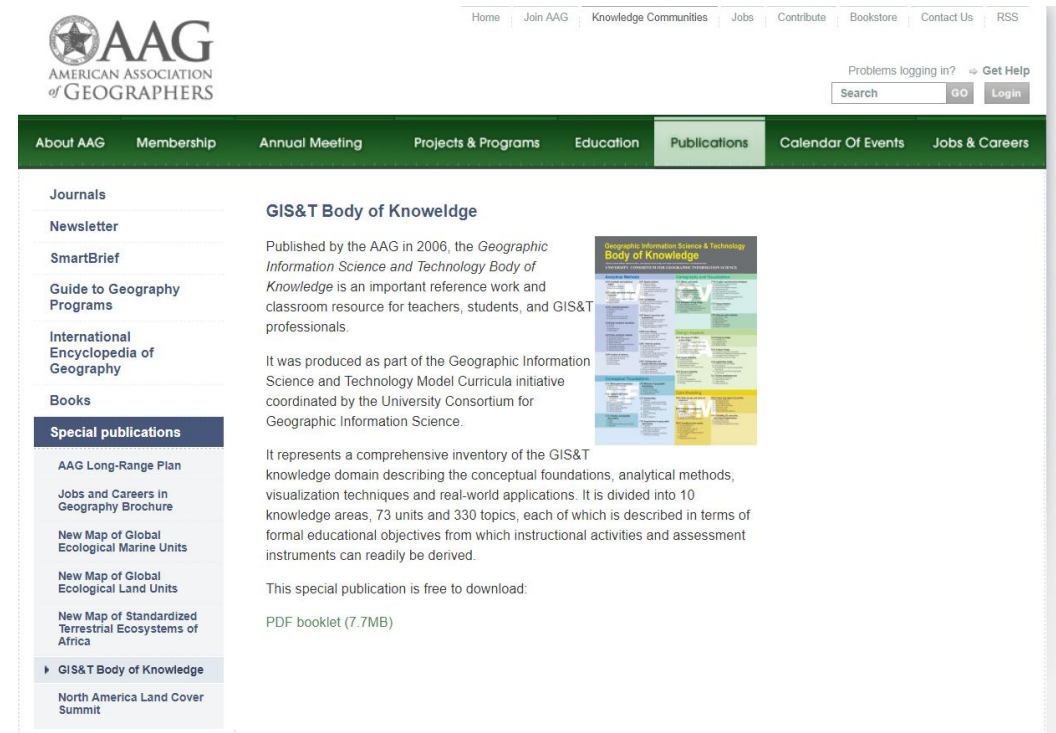
2022

1. Two current students worked as part-time at ARDOT



Geospatial Skillsets

1. Spatial problem solving
2. Spatial analysis
3. Software troubleshooting
4. Statistics
5. Spatial predictive modeling
6. Database management
7. Web GIS
8. Python scripting
9. Cartography/Map design
10. Graphic design
11. Image interpretation
12. GPS tracking



Esri Summer Internship

12-WEEK SUMMER INTERNSHIP

Innovation and collaboration



Real work. Real experience.

Work on projects that matter, such as creating an app to speed emergency aid or conducting research for a marketing campaign.



Work hard and play hard

Participate in daily events and programs, including Monday Meetups, Tech Tuesday, What's Next Wednesday, Training Thursdays, and Fun Friday.



Accelerated personal growth

One-on-one mentoring, professional development workshops, and various networking opportunities provide valuable tools to jump-start your career.



Esri Intern Hackathon

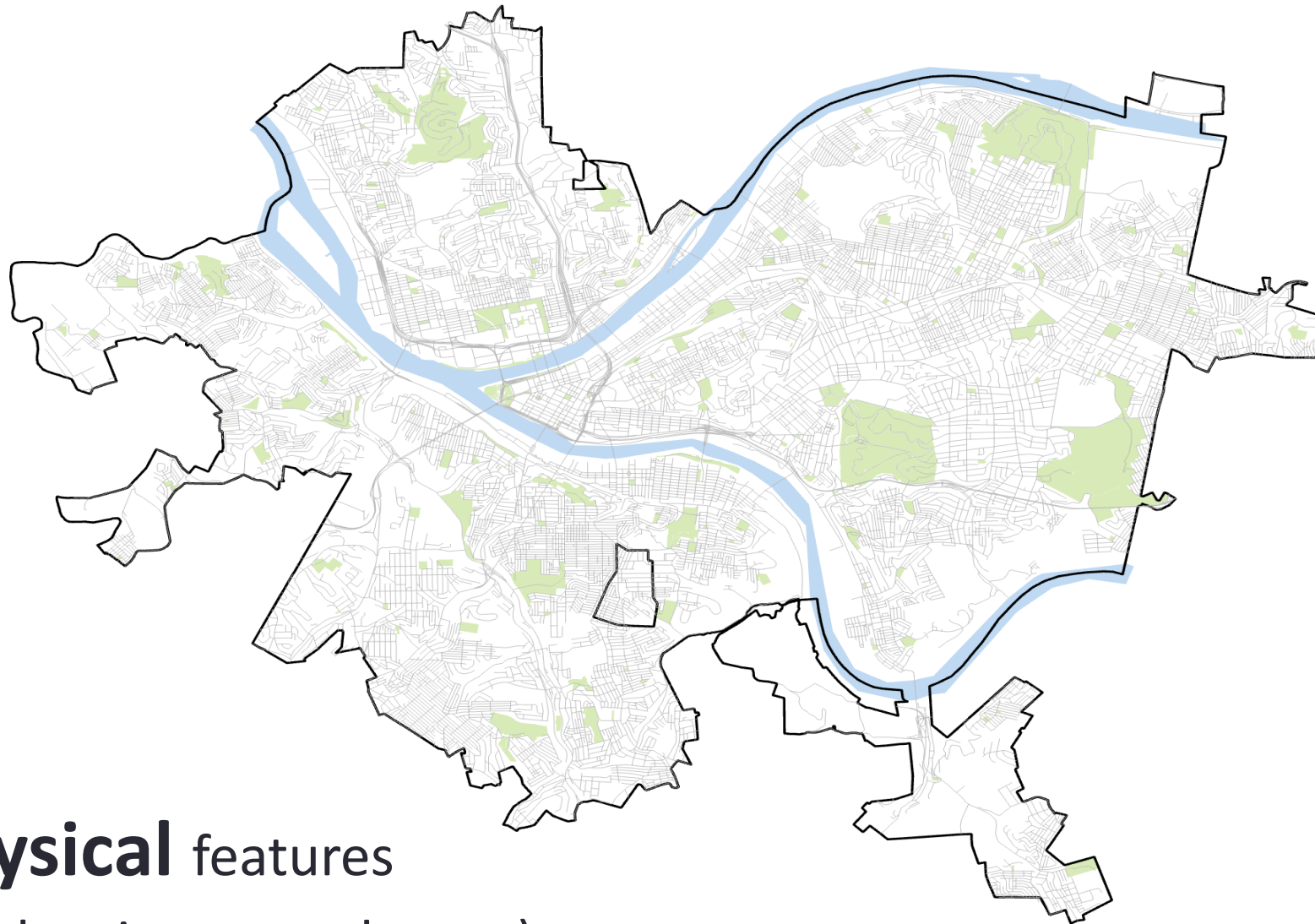
2.2

FEATURE CLASSES AND MAP LAYERS

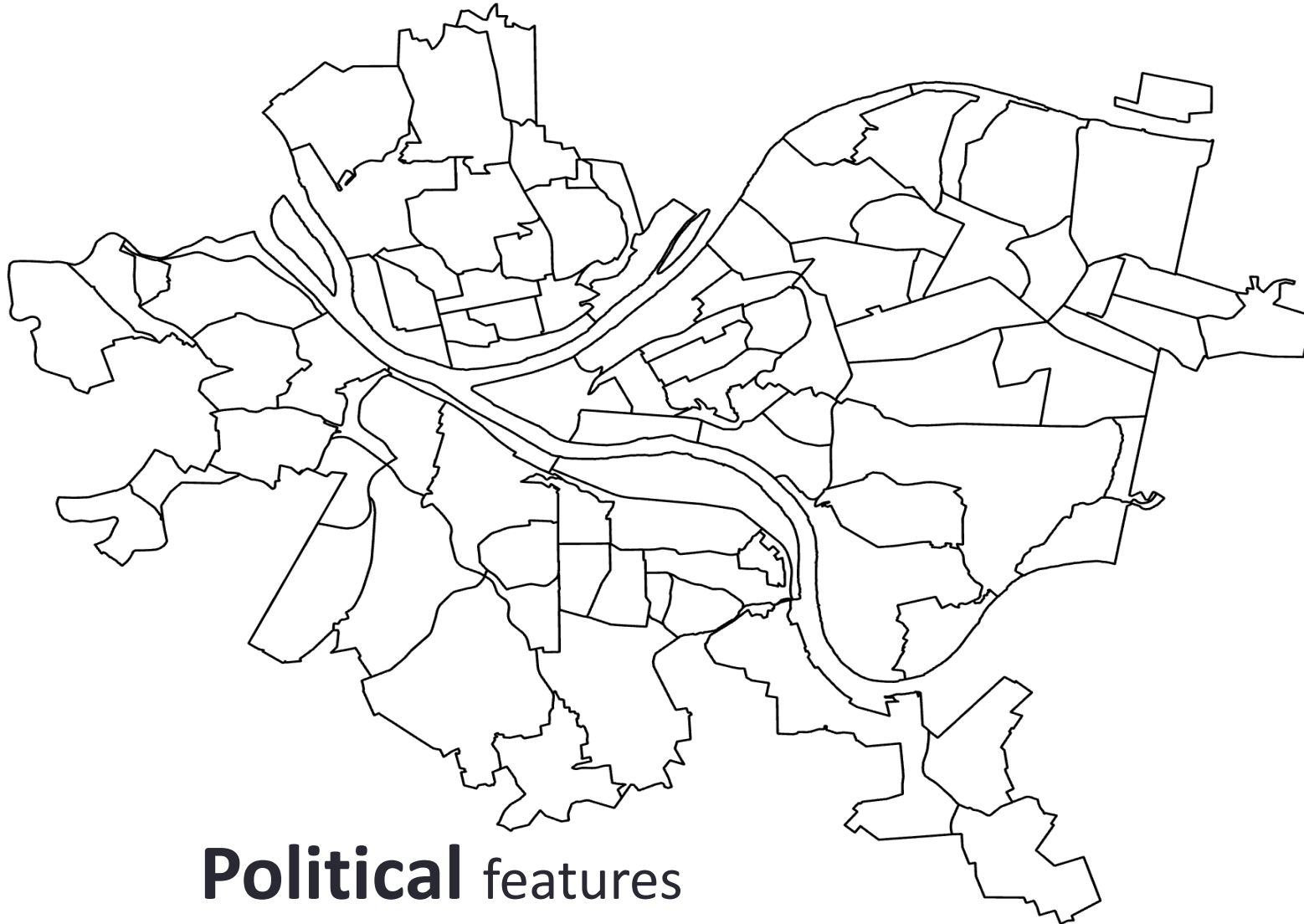
GIS stores related geographic features as separate collections of files called **feature classes**

1. can be easily reused, assembled into any number of map compositions, and overlaid for analysis
2. are associated with attribute tables of data

When symbolized and displayed as part of a map, a feature class is called a **map layer**



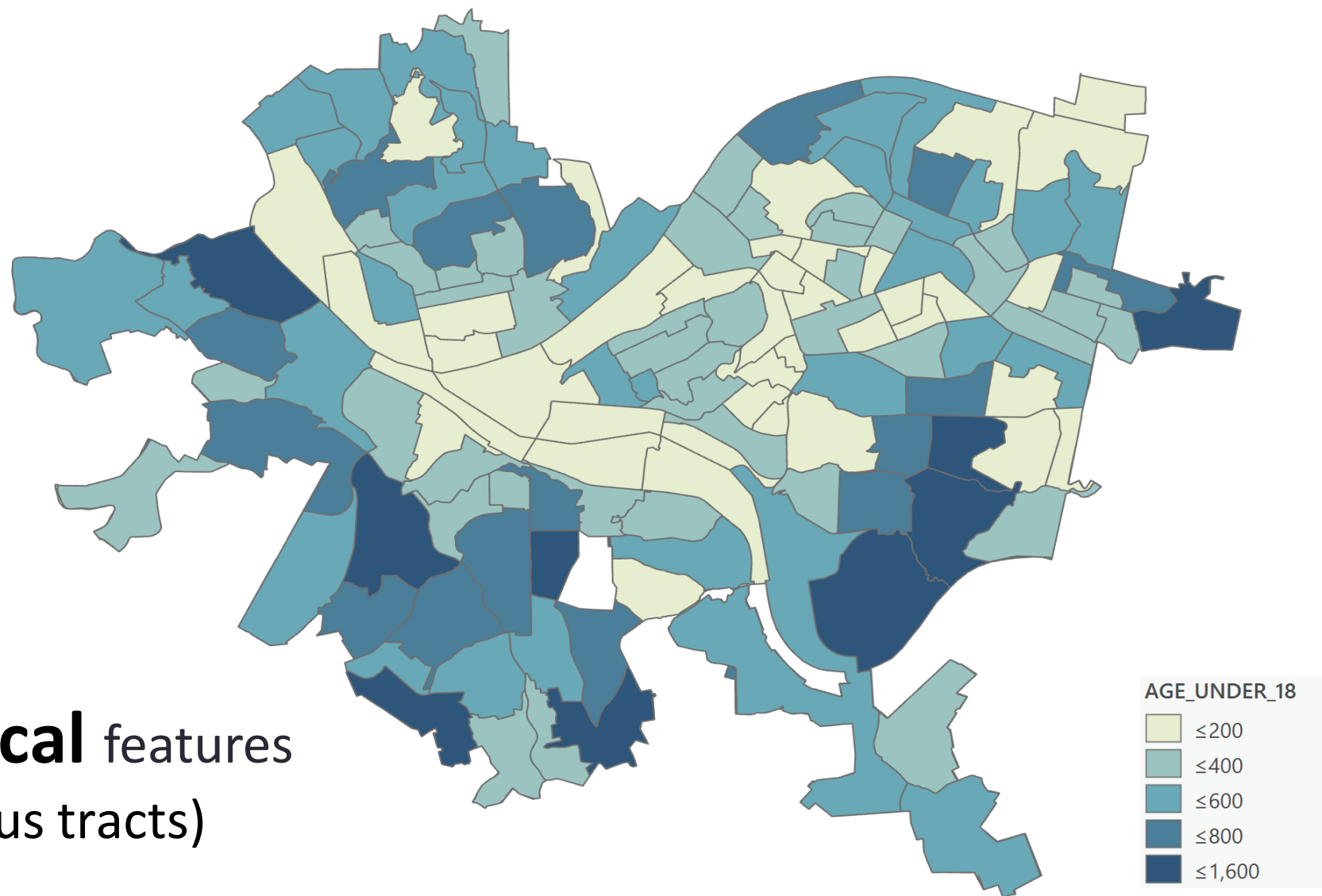
Physical features
(lakes, parks, rivers, roads, etc.)

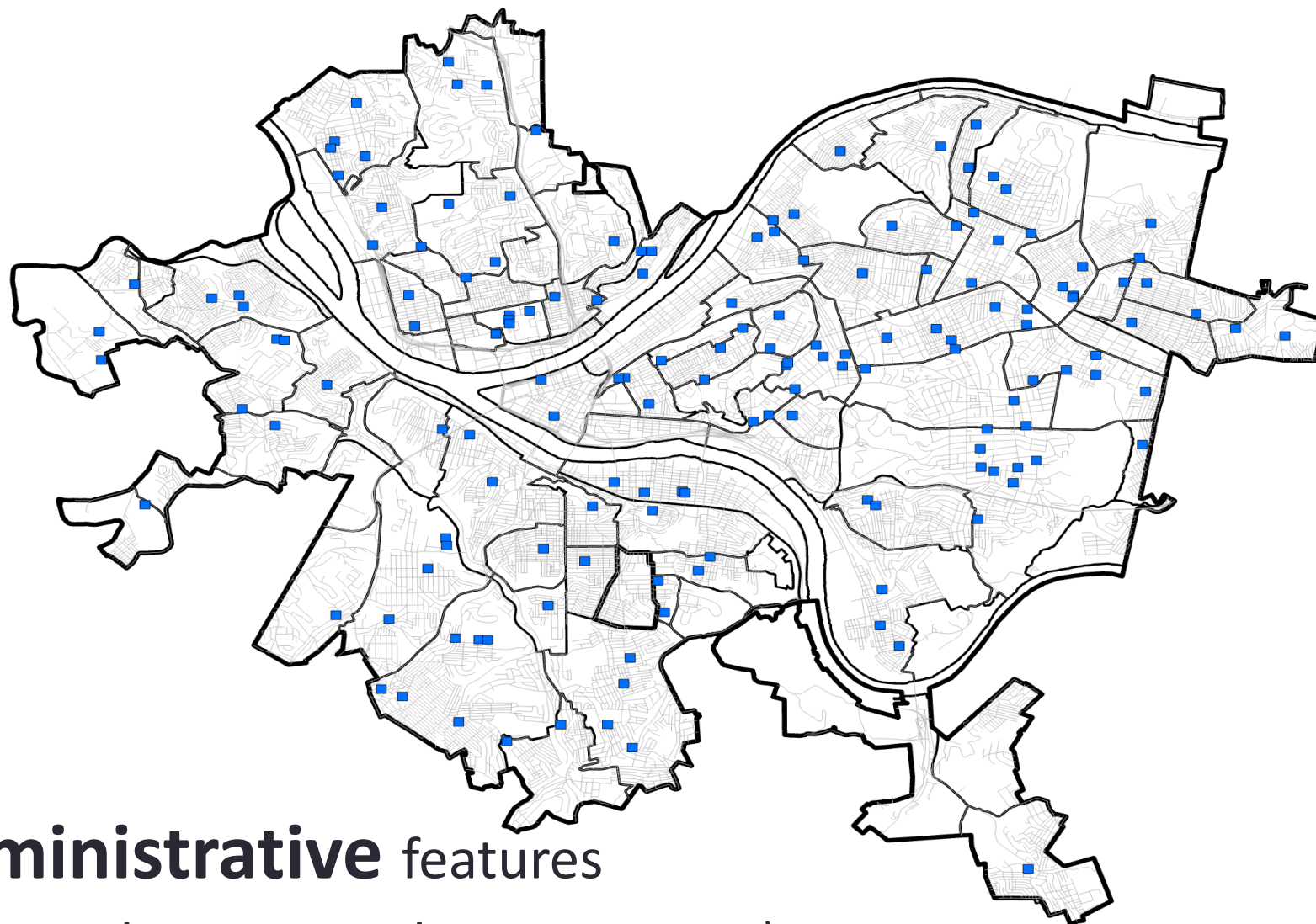


Political features

(states, provinces, municipalities, neighborhoods)

Statistical features (Census tracts)

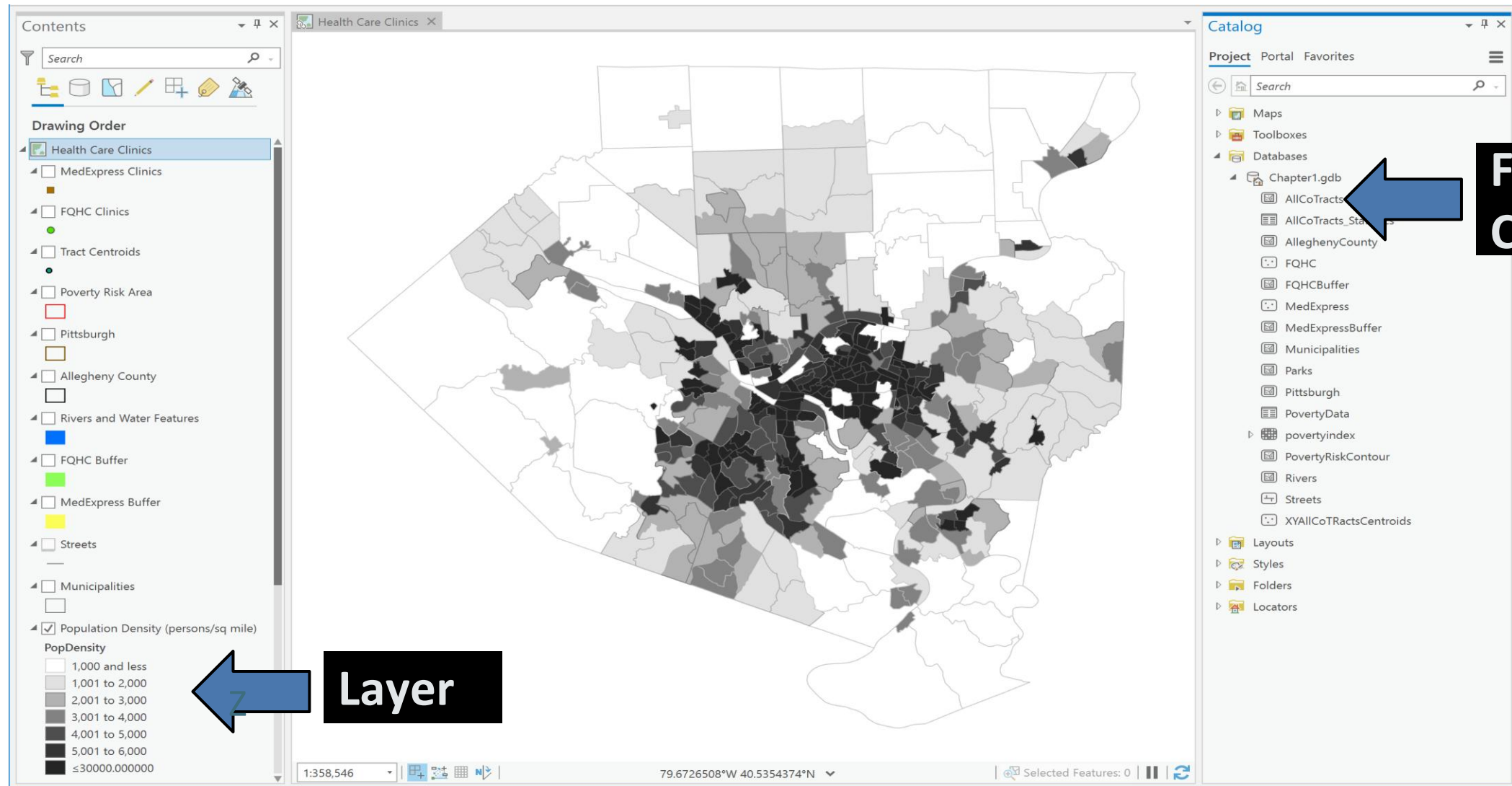




Administrative features

(schools, clinics, businesses, bus stops, etc.)

Layers and features as viewed in ArcGIS Pro



2.3

ARCGIS ONLINE

About Content Legend

Legend

MedExpress Clinics

FQHC Clinics

Pittsburgh

Allegheny County

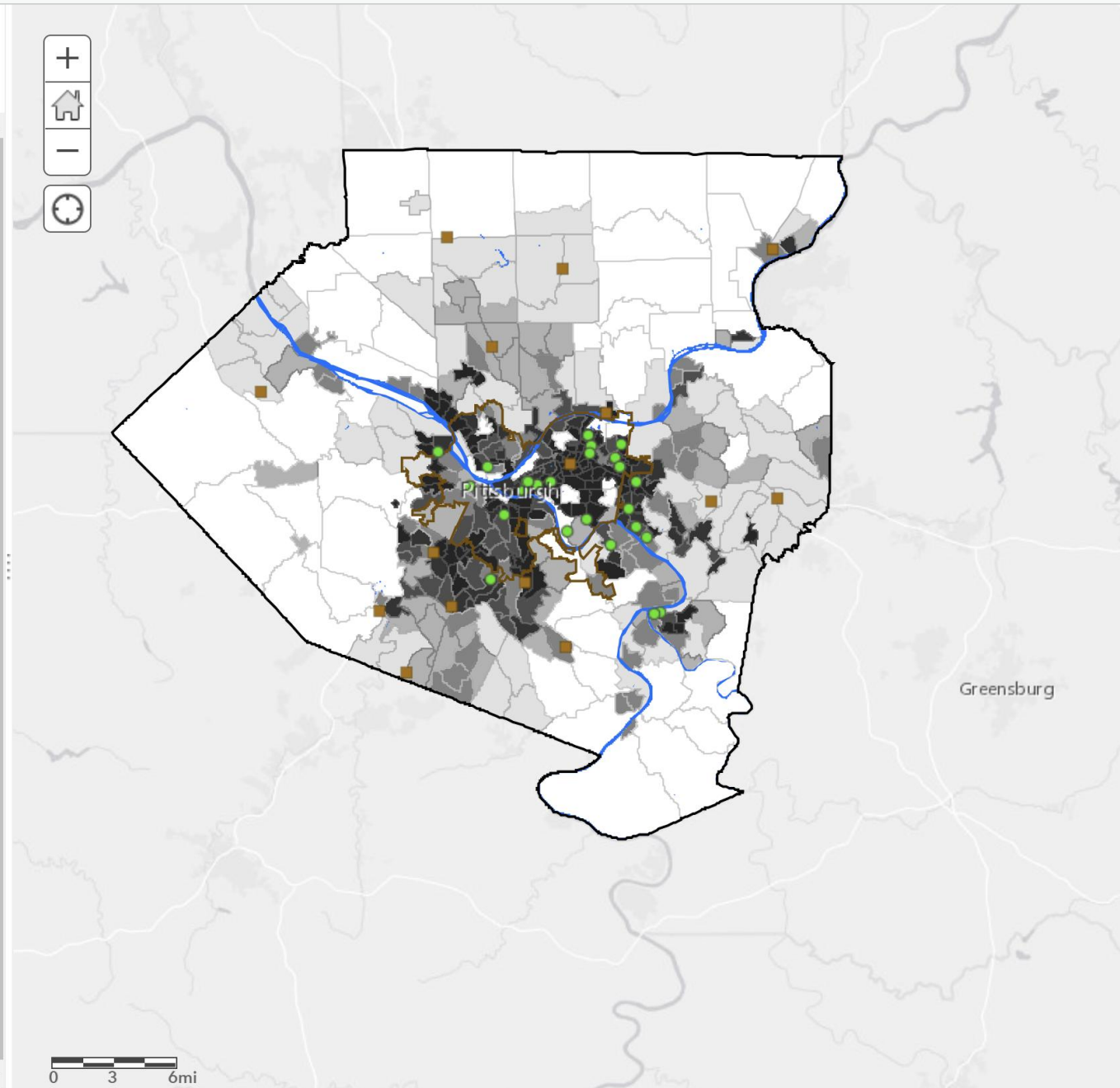
Rivers and Water Features

Population Density (persons/sq mile)

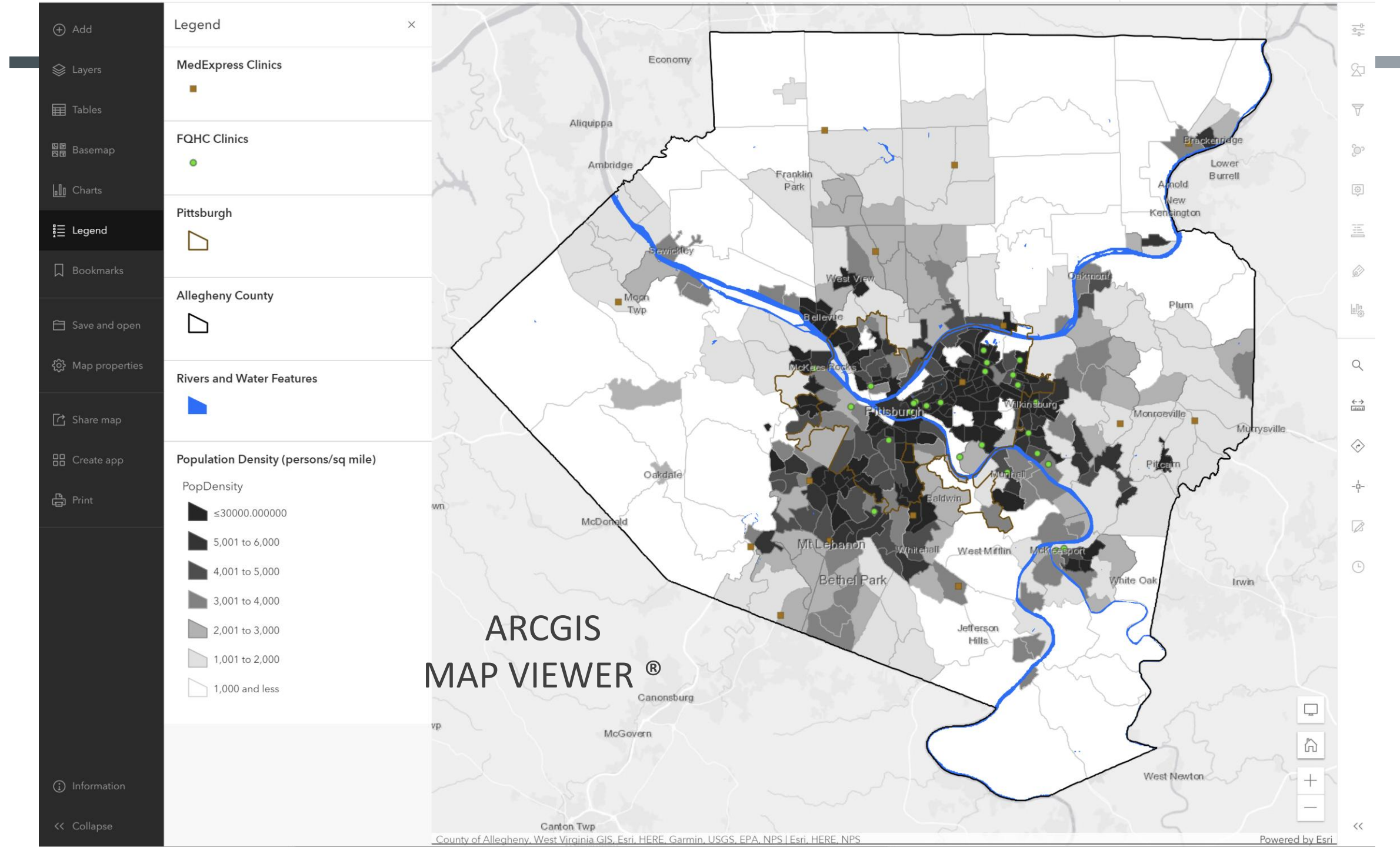
PopDensity

≤30000.000000
5,001 to 6,000
4,001 to 5,000
3,001 to 4,000
2,001 to 3,000
1,001 to 2,000
1,000 and less

Trust Center Contact Esri Report Abuse Contact Us



ARCGIS ONLINE[®]
(MAP VIEWER
CLASSIC)



Arts employment by state and metropolitan area

Employment levels, average wages, and cost of
living index

Mary Smith
Draft

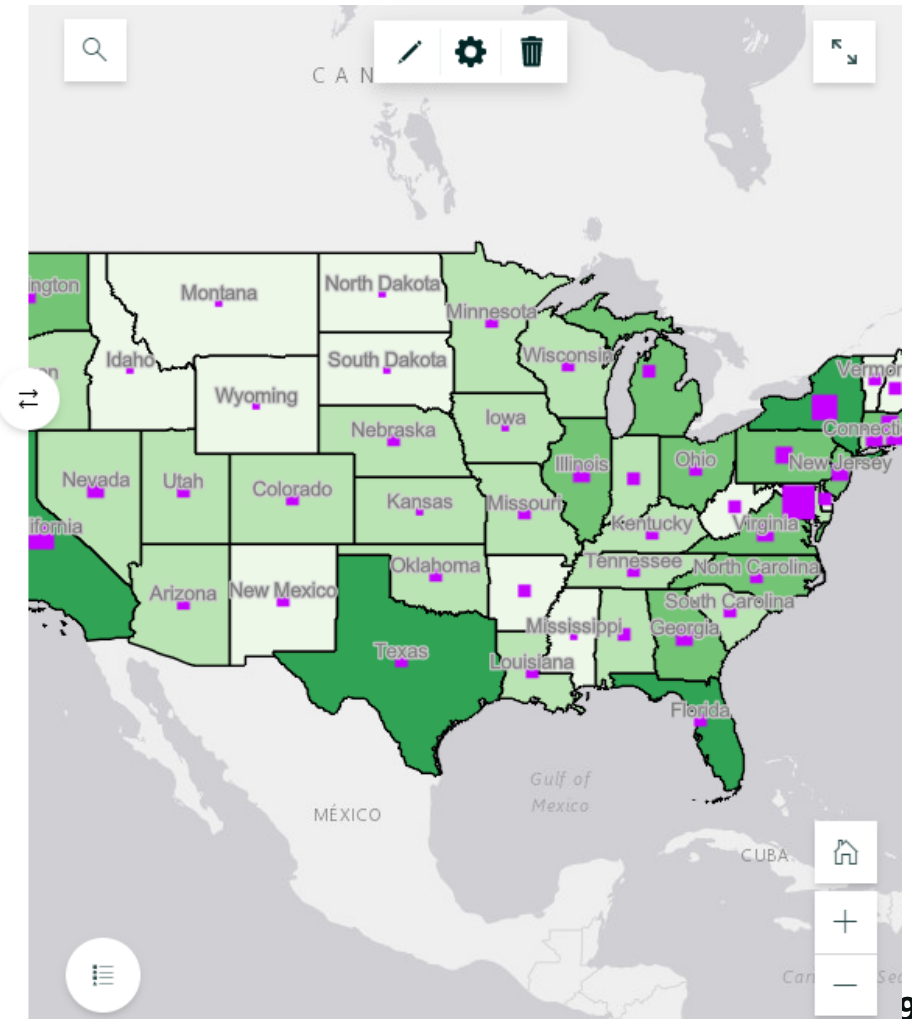


ArcGIS Story Maps®

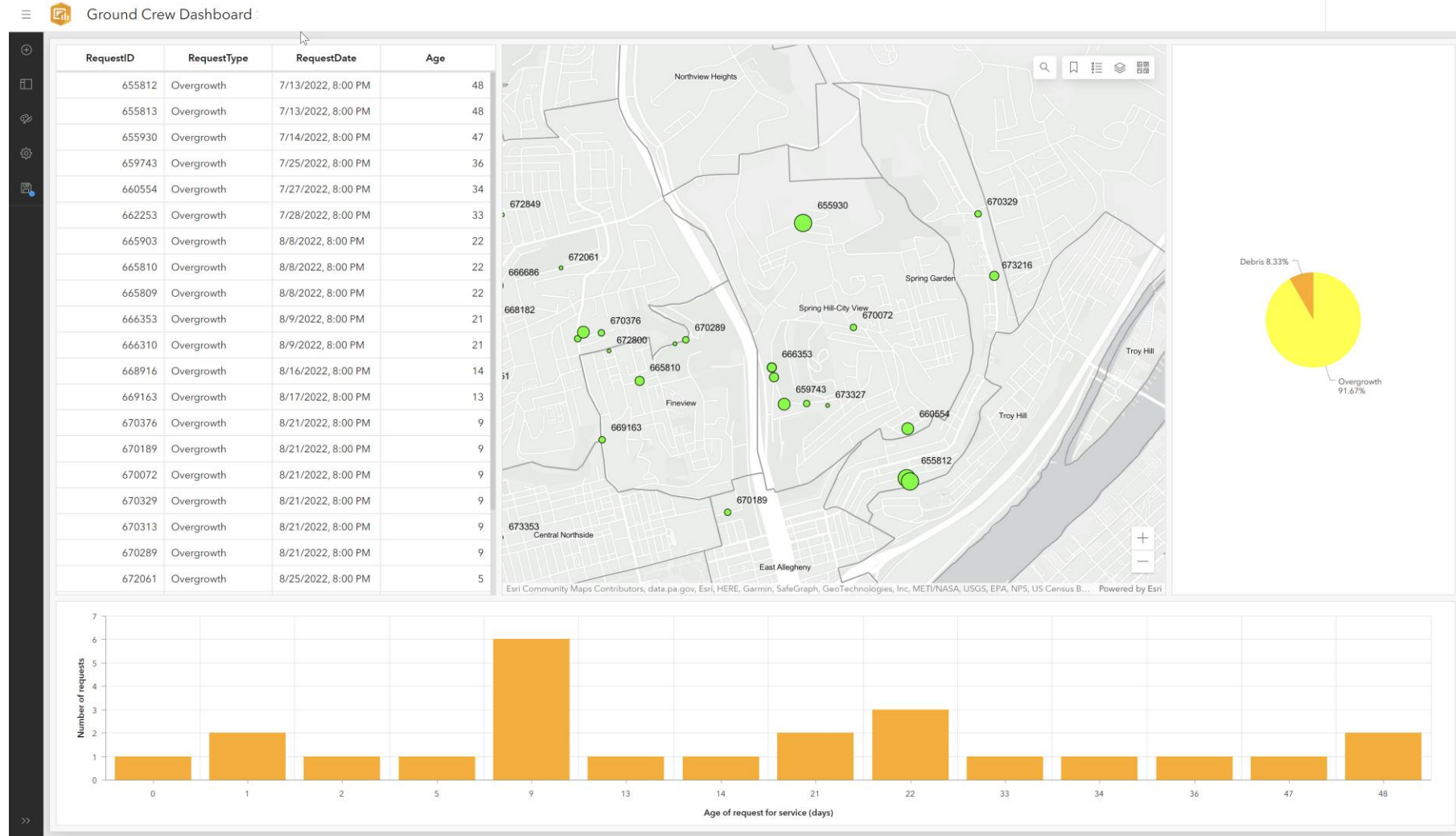
Arts Employment map

Map use and navigation

- Hover over the legend icon (lower left of map) to see the map legend.
- Click the magnifying glass icon (upper left of map) to search for an address or place.
- Click the + to zoom in, - to zoom out, and house icon to zoom to the whole map.
- Zoom in far enough and



ArcGIS Dashboard®





2.4

LAB SESSION

LAB 01: INTRODUCTION TO ARCGIS PRO



DUE BY MIDNIGHT
(11:59 PM) ON
MONDAY, SEP 1, 2025



SUBMIT A WORD
DOCUMENT ONLY INCLUDE
QUESTION AND ANSWER